

G S Lahari

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SUMMARY

Data and AI Engineer with 2+ years building production pipelines and automation systems at Caterpillar Inc. I own Snowflake data pipelines for the Product Support & Logistics Division, built Power Automate and Copilot AI workflows that manage real-time warehouse exception tracking across 20+ global facilities, and was selected for a month-long on-site deployment at Caterpillar's US headquarters. Outside work, I build LLM observability and prompt evaluation tooling. I work at the intersection of data engineering and applied AI — building systems that make both the data layer and the AI layer reliable, measurable, and production-ready.

EXPERIENCE

Caterpillar Inc.

Bengaluru, India

Data & AI Automation Engineer (IT Analyst, Applications I)

Mar 2025 – Present

- Built a Power Automate workflow that pulls live exception data from a Power BI monitoring dashboard, creates and manages parent-child records across two SharePoint lists, fires structured Teams alerts when new handling unit discrepancies appear, and automatically progresses status from NEW to IN PROCESS to COMPLETED — **eliminating duplicate alerts and manual tracking** for warehouse ops teams across **20+ global facilities**.
- Built 5+ additional AI automation workflows and Copilot agents handling routing, triage, and reporting tasks — **reducing recurring manual operations effort by 300+ task executions per week** across warehouse operations teams.
- Selected for a month-long on-site deployment at Caterpillar's US headquarters to stabilise AWCS (Automated Warehouse Control Systems), the core platform managing HQ warehouse operations — diagnosing and resolving production failures directly in C, C++, and Java codebases, including a critical buffer-handling defect that had been causing intermittent data corruption across downstream systems.
- Owned production reliability for the AWCS stack, resolving **500+ incidents annually** by tracing root causes in code rather than patching tickets — reducing repeat incident rate by an estimated **30%** and improving system uptime across HQ operations.
- Led cross-functional delivery of automation features from requirements through production across engineering, infrastructure, and operations teams in the US, India, and Europe.

Data Engineer, PS�D | CArE (Systems Data Support Analyst)

Mar 2024 – Mar 2025

- Designed and built multi-layer Snowflake analytics tables and ETL pipelines feeding supply chain KPIs across the Product Support & Logistics Division — owning source-to-consumption data flow including ingestion, transformation logic, data modeling, and pipeline scheduling for daily batch loads consumed by Tableau and Power BI reports across multiple regions.
- Implemented automated data quality checks across production pipelines — including null validation, row count reconciliation, referential integrity checks, and threshold-based anomaly detection with automated alerting — **cutting data discrepancy incidents by 40%** and making downstream reporting reliable enough for senior leadership consumption.
- Owned incident response for **50+ pipeline failures per quarter** with a sub-24-hour SLA, tracing root causes across source systems, transformation layers, and downstream consumers — reducing time-to-resolution through structured runbooks built from recurring failure patterns.
- Authored full pipeline documentation covering data dictionaries, transformation logic, dependency maps, and runbooks — cutting new-joiner ramp time by approximately two weeks and eliminating single-point-of-knowledge risk on critical pipelines.

Amazon

Chennai, India

Quality Assurance Engineer, Intern

Jan 2023 – Jun 2023

- Built an internal automation tool that scraped open defects from the bug tracking system and delivered structured weekly digest emails to the QA team — **saving 3 hours of manual tracking per week** and becoming the team's standard defect visibility process.
- Extended the Diamond Tool Kit automation framework with new POJO classes for Multi-Tender BNPL payment pipelines across Saudi Arabia and UAE, **increasing regression coverage by 25%** on high-value transactional flows.
- Wrote and executed 100+ test scenarios on high-value payment flows, joined bug triage sessions, and contributed defect-pattern analysis that reduced pre-launch defect counts across two markets.

PROJECTS

Prompt Canary | *Python, FastAPI, GitHub Actions, SQLite, Streamlit, Slack API* | github.com/GSLAHARI/Prompt.Canary2026

- Built a CI/CD regression gate for LLM-powered features that **automatically blocks PR merges** when a prompt change causes category accuracy to drop more than 8% or LLM-as-judge score to drop more than 0.5 — treating prompts as versioned, testable code with the same safety guarantees as application code.
- Designed a **60-case golden evaluation dataset** across easy, medium, hard, and adversarial difficulty levels — including prompt injection attempts, multilingual inputs, and deliberately ambiguous edge cases — with 7-run rolling-average drift detection to catch gradual degradation that point-in-time diffs miss.

- Runs entirely in mock mode at **zero API cost**; CI triggers on **prompts/**** changes, generates self-contained HTML diff reports with inline SVG trend charts, and posts structured Slack Block Kit scorecards directly on the PR.
- LLM Sentinel** | *Python, FastAPI, Pydantic v2, SQLite, Streamlit, Docker, GitHub Actions* | github.com/GSLAHARI/llm-sentinel2026
- Built a production observability gateway that intercepts every LLM API call — logging requests, latency, cost, and quality scores — giving teams full incident-response visibility over their LLM stack, the same way APM tools cover application services.
 - Implemented a **cost-aware model router** that selects the cheapest model meeting a configurable quality threshold and escalates for complex prompts, alongside **4 anomaly detectors** with tuned thresholds: rolling P95 latency spikes, per-feature daily budget overruns, quality score drops, and error rate surges — each firing Slack Block Kit alerts at warn and critical levels.
 - Fully containerised with Docker Compose; GitHub Actions CI runs **20 smoke tests in mock mode** on every push, with a traffic simulator and anomaly injection scripts for load testing and demo validation.
- Snowflake & dbt RAG System** | *Python, Qdrant, BM25, RRF, RAGAS, Claude Sonnet, Langfuse, Docker* | [github.com/GSLAHARI/s](https://github.com/GSLAHARI/snowflake-rag)
- Built a Q&A system over Snowflake and dbt documentation using **hybrid retrieval** — BGE-large dense vectors merged with BM25 via Reciprocal Rank Fusion (k=60), reranked by a BGE cross-encoder — choosing this architecture specifically because vector-only retrieval fails on the precision-sensitive facts and multi-document synthesis questions that matter most in this corpus.
 - Hand-curated a **50-question benchmark** across factual, conceptual, architectural, and adversarial tiers with deliberate precision traps (Time Travel vs Fail-safe, exactly-once vs at-least-once task delivery, dbt as orchestrator misconception) to stress-test retrieval accuracy; ran 3-config ablation studies achieving **0.86 faithfulness** on conceptual questions in the best configuration.
 - CI gate fails the build when RAGAS faithfulness drops below 0.80; all queries traced end-to-end in Langfuse. Documented 6 root-cause failure cases — including query rewrite misdirection and corpus scope gaps — with specific remediations applied and re-evaluated.

TECHNICAL SKILLS

Data Engineering: Snowflake (data modeling, ETL/ELT, incremental loads, query optimization), dbt, Apache Airflow, Data Quality Frameworks (null checks, row reconciliation, anomaly detection), Pipeline Observability, SnapLogic

AI & LLM Engineering: RAG, Hybrid Retrieval (BM25 + RRF), Cross-Encoder Reranking, Prompt Regression Testing, LLM Observability, RAGAS, Langfuse, LangChain, Azure OpenAI

Automation Engineering: Power Automate (status lifecycle, deduplication, SharePoint/Teams integration), Microsoft Copilot Studio, FastAPI, Workflow Automation, GitHub Actions CI/CD

Languages: Python, SQL, Java, C, C++, JavaScript

Backend & Infra: FastAPI, Pydantic v2, Docker, REST APIs, SQLite

Databases & Vector Stores: Qdrant, Snowflake, MySQL, SQL Server, Oracle

Visualisation & BI: Power BI (DAX, Power Query), Tableau, Streamlit

Cloud: Microsoft Azure (AZ-900) **Tooling:** Git, Jupyter, Agile/Scrum

CERTIFICATIONS

Microsoft Azure Fundamentals, AZ-900 | Microsoft

IBM Data Science Professional Certificate | Coursera

EDUCATION

K L University
Bachelor of Technology, Computer Science and Engineering

Vijayawada, AP
Jun 2019 – May 2023